NOTES

TUT 7

1) Josepher's Problem.

→

Compute for N = 16

N	1	2	3	4	5	6	7	8	9
J(N)	1	1	3	1	3	5	7	1	3

10	11	12	13	14
5	7	9	11	13

15	16
15	1

each time we get to 2^n J(N) resetes

Base case

$J(1) = 1$

Recursive

$J(2n) = 2.J(n) - 1$ (even)

$J(2n+1) = 2.J(n) + 1$ (odd)

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Algorithm

```
funct^n Josepher's
    If n=1
        return 1
    If n = even
        return 2* Josepher's (n/2) - 1
else
    return 2* Josepher's [(n-1)/2] + 1
```

Time complexity ($O[\log n]$).

Write N in form

$N = 2^m + L$ where

$2^m \leq N < 2^{(m+1)}$

& ~~L = N~~ $L = N - 2^m$.

Then survivor's position is

$$\boxed{J(N) = 2L + 1}$$

NOTES

Verify

N	Binary	2^M	$L = N - 2^M$	J(N)
1	1	1	0	
2	10	2	0	1
3	11	2	1	1
4	100	4	0	3
5	101	4	1	1
6	110	4	2	3
7	111	4	3	5
8	1000	8	0	7
9	1001	8	1	1
10	1010	8	2	3
11	1011	8	3	5
12	1100	8	4	7
13	1101	8	5	9
14	1110	8	6	11
15	1111	8	7	13
16	1000	16	0	15
				1

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2.

→ Divide & conquer

Euclidean distance between

$a = (x_1, y_1)$ $b = (x_2, y_2)$ is

$$d(a,b) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Brute force all $(n/2) = n(n-1)/2$.

gives $O(n^2)$ complexity.

Before recursion

funct^n ClosestPair (P):

$P_x \leftarrow P$ sorted by x-coordinate

$P_y \leftarrow P$ sorted by y-coordinate

Return Closest_Rec (P_x, P_y)

NOTES

Main recursive

funct^n ClosestPair_Rec (Px, Py)

$n \leftarrow |Px|$

If $n \leq 3$;

Return Brute force (Px)

mid $\leftarrow$ n/2

midPt $\leftarrow$ Px[mid]

Lx $\leftarrow$ Px[1 .... mid]

Rx $\leftarrow$ Px [mid+1 ... n]

Ly, Ry $\leftarrow$ Split Py based on

whether $x \leq$ midPt.x

or

$x \geq$ midpt.x

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$\delta_L \leftarrow$ Closest_Rec(L_x, L_y)

$\delta_R \leftarrow$ Closest_Rec(R_x, R_y)

$\delta \leftarrow \min(\delta_L, \delta_R)$

Return min (δ, StripClosest (P_x, P_y, midpt, δ))

3)

- Given integer $A[1,..n]$ for each $A[i]$ find count of all elements $A[j]$ such $j > i$ & $A[j] < A[i]$

→ Modified Merge Sort

Relation

$T(n) = 2T(n/2) + O(n)$

By Master theorm

Case 2 →

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$T(n) = O(n \log n)$

Algorithm

Funct^n count (A[1.... n])

counter [1... n]

MergeSort (A, 1, n, counter)

Return counter.

Funct^n MergeSort (A, left, right, counter)

If left > right

return

mid ← (left + right)/2

Merge Sort (A, Left, mid, counter)

Merge Sort (A, mid+1, right, counter)

Merge ~~Sort~~ (A, right, left, mid, counter)

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```
Function Merge(A, left, right, mid, counter)
    L ← A[Left .. mid]
    R ← A[mid+1 ... right]
    i ← 0, j ← 0, k ← left
    right count ← 0
while i < len(L) & j ← len(R)
    if L[i] ≤ R[j]:
    count [original index of L[i]] += right counter
        A[k] ← L[i]
        i ← i+1
else
    right Counter ← right Counter + 1
        A[k] ← R[j]
        j ← j+1
        k ← k+1
```

NOTES

While $i < Len[L]$;

counter [orignial index of $L[i]$] +=

While $j < len(R)$;

$A[k] \leftarrow R[j]$

$j \leftarrow j+1$, $k \leftarrow k+1$.